"ADVANCED CRUISE CONTROL"

Project Report

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Submitted in partial fulfillment of the requirements for the award of

Bachelor of Engineering

In

Electronics and Communication Engineering

Submitted to

VisvesvarayaTechnological University

Belagavi, Karnataka, 590 014

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2024-2025

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CERTIFICATE

Certified that the project work entitled "**Advanced Cruise Control**" is a bonafide work carried out by **Abhishek A R (2KE21EC003)**, **Ashish Khot (2KE21EC016)**, **Govindagouda P (2KE21EC026)**, and **Manish R M (2KE21EC041)**, in partial fulfilment for the award of degree of Bachelor of Engineering in VII Semester, Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi, during the year 2024-25. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the said degree.

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DECLARATION

We, **Abhishek A R (2KE21EC003), Ashish Khot (2KE21EC016), Govindagouda P (2KE21EC026), and Manish R M (2KE21EC041)** students of VII Semester B.E., K.L.E. Institute of Technology, Hubballi, hereby declare that the Project has been carried out by us and submitted in partial fulfillment of the requirements for the VII Semester degree of **Bachelor of Engineering in Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi** during academic year 2024- 2025.

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This is to certify that the project work entitled **"Advanced Cruise Control"**, is carried out by **Abhishek A R (2KE21EC003)**, **Ashish Khot (2KE21EC016)**, **Govinda gouda P (2KE21EC026)**, and **Manish R Marutagi (2KE21EC041)**, students under my direct supervision, for the award of the degree of Bachelor of Engineering in **"Electronics and Communication Engineering"** of VTU, Belagavi, Karnataka.

To the best of my/our knowledge and belief,

(i) The work is carried out by the candidates only.

(ii) The work is not taken in its original form, from any other thesis or project reports.

(iii) The work is up to the desired standard both in the technical content and the write up.

(iv) Plagiarism percentage is 24% .

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The project work as mentioned above is hereby recommended and forwarded for examination and evaluation.

Dr. Gopal Bidkar

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ABSTRACT

Humps or speed breakers play a vital role in traffic management and road safety. Detecting and accurately identifying humps in real-time is crucial for autonomous vehicles, advanced driver- assistance systems, and road maintenance authorities. The proposed methodology involves training a YOLOv8 model on a large dataset of annotated images, specifically focused on hump detection. The dataset comprises diverse road scenarios, encompassing different lighting conditions, weather conditions, and hump designs. By leveraging the YOLOv8 architecture, the model is capable of detecting humps in real-time with high accuracy and efficiency. Experimental results demonstrate the efficacy of the proposed approach. Additionally, the hump height is estimated based on the detected hump's dimensions. This information is used to dynamically adjust the vehicle's speed through an integrated speed control system. The speed control mechanism considers the hump's height and applies appropriate speed reduction strategies. Real-world testing in different road environments validates the system's robustness and adaptability.

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LIST OF ABBREVIATIONS

Abbreviation	Description
ACC	Adaptive Cruise Control
ROI	Region of interest
ADAS	Advanced Driver Assistance Systems
EMS	Energy Management Strategy
HEV	Hybrid Electric Vehicle
CACC	Cooperative Adaptive Cruise Control
ITS	Intelligent Transportation System
ESC	Electronic Stability Control
RADAR	Radio Detection and Ranging
LCD	Liquid Crystal Display

CHAPTER 1

INTRODUCTION

Controlling the speed of a vehicle according to the hump height in cruise mode is a novel approach that combines the convenience of cruise control with intelligent speed adjustment based on hump characteristics. Humps, or speed breakers, are road features designed to regulate vehicle speed in specific areas. However, abrupt or poorly designed humps can cause discomfort, vehicle damage, or even accidents if approached at excessive speeds.

Cruise control systems are widely used in modern vehicles to maintain a constant speed without continuous driver intervention. These systems allow drivers to set a desired speed, which the vehicle then maintains automatically. However, cruise control traditionally lacks the ability to detect and adapt to road obstacles, such as humps. By integrating hump detection and speed control mechanisms into cruise mode, vehicles can automatically adjust their speed based on hump height, providing an added layer of safety and comfort.

The integration of hump height-based speed control in cruise mode offers numerous benefits. Firstly, it enhances safety by preventing accidents caused by approaching humps at excessive speeds. The system ensures that the vehicle's speed is automatically adjusted to a level suitable for traversing the hump safely. Secondly, it improves passenger comfort by providing a smoother ride over humps, minimizing discomfort and jolts experienced by occupants. Moreover, it helps prevent potential damage to the vehicle's suspension system by reducing the impact and stress exerted on the vehicle when encountering humps.

As automotive technology continues to advance, future developments may include adaptive cruise control systems that incorporate hump detection and speed adjustment as an integrated feature. This integration would provide drivers with a seamless and automated driving experience, where the vehicle dynamically adjusts its speed in response to encountered humps while in cruise mode.

1.2 LITERATURE REVIEW AND RESEARCH GAP

1. A Study on Adaptive Cruise Control, International Research Journal of Engineering and Technology (IRJET) Volume: 08 Issue: 10 | Oct 2023

The first generation of adaptive cruise control (ACC)-equipped vehicles hit the market 15 years ago, and the ISO standard for the first generation of ACC systems was published 7 years ago. Because the next generation of ACC systems and more advanced driver-assistant systems are about to be fully introduced and deployed, it's important to review the development and research achievements of the first generation of ACC systems in order to provide more practical experience for the new deployment. This study examines related development and research achievements from a variety of angles in order to objectively and thoroughly introduce them an ACC system to researchers, automakers, governments and consumers. It makes an attempt to describe what an ACC system is and how it works in a systematic manner.

This article presents a small-scale testbed for automated driving. The main benefits of small-scale testing are safety, affordable hardware, and independence of traffic regulations and test tracks. The testbed using model trucks (scale 1:14) and model cars (scale 1:10) is described in detail and exemplary results using different advanced driver assistance systems are presented. The focus in this work lies on control engineering topics such as trajectory planning, tracking, prediction and estimation algorithms.

The automation of road vehicles implies highest demands for all parts involved in the driving task, including the vehicle actuators. Fault-tolerant trajectory tracking, which exploits functional redundancies among the vehicle actuators, bears the potential to avoid fail-operational requirements on actuator level. Yet, the use of fault-tolerant trajectory tracking must be well argued as the number of possible driving scenarios of automated vehicles is virtually unlimited. We present a reference trajectory generation approach in order to support such a safety argumentation. The generation essentially respects functional limits of the specific vehicle automation application, e.g., maximum speed or lateral acceleration. Derived from standard maneuvers of automated vehicles, the trajectories are generated with varying paths and speed profiles for the specific application. Eventually, a set of trajectories is generated for an example application and applied to investigate the potential of a selected fault-tolerant trajectory tracking approach. The simulation results for the example application reveal the shortcomings of the selected fault-tolerant trajectory tracking algorithm. However, these results also indicate that fault-tolerant trajectory tracking can be employed within the example application once the shortcomings of the algorithm are resolved.

2. Researches on Adaptive Cruise Control system: A state-of-the-art review, Volume: 236 Issue: May 26, 2024

Adaptive Cruise Control (ACC) is one of Advanced Driver Assistance Systems (ADAS) which takes over vehicle longitudinal control under necessary driving scenarios. Vehicle in ACC mode automatically adjusts speed to follow the preceding vehicle based on evaluation of the surrounding traffic. ACC reduces drivers' workload as well as improves driving safety, energy economy, and traffic flow. This article provides a comprehensive review of the researches on ACC. Firstly, an overview of ACC controller and applied control theories are introduced. Their principles and performances are discussed. Secondly, several application cases of ACC control algorithms are presented. Then validation work including simulation, Hardware-in-the-Loop (Hil) test and on-road experiment is descripted to provide ideas for testing ACC systems for different aims and fidelities. In addition, studies on human-machine interaction are also summarized in this review to provide insights on development of ACC from the perspective of users. At last, challenges and pot1

ential directions in this field is discussed, including consideration of vehicle dynamics properties, contradiction between algorithm performance and computation as well as integration of ACC to other intelligent functions on vehicles.

3. Adaptive Cruise Control Strategies Implemented on Experimental Vehicles: A Review, IFAC-Papers On Line Volume: 52, Issue :5, 2023

It has been over two decades since the first generation of adaptive cruise control (ACC) equipped vehicles were launched onto the market. However, control strategies adopted by commercially available ACC systems are the closely-guarded intellectual property of their industrial developers, so these are not publicly available and there is no uniform standard for ACC dynamic response. However, the impact of ACC dynamics on transport networks needs to be studied. In the open literature, there are many ACC systems published and embedded in simulation studies, nevertheless, inappropriate model design can easily point to misleading conclusions. Only a few projects dealing with automated systems are founded on real experimental data, which is important for developing precise ACC models and furthermore, evaluating their realistic effects on highway capacity and traffic flow dynamics. This review firstly summarizes the available literature on control strategies for ACC systems implemented on experimental vehicles, then propose a five-layer framework for the development and

Optimization methods for Energy Management Strategy (EMS) are more and more used to minimize the fuel consumption of Hybrid Electric Vehicles (HEV). For off-line techniques, such as dynamic programming, from a given driving cycle, an algorithm is defined to determine the optimal power flows of the vehicle. Most of the time, the optimization is achieved by not considering the resistive force due to the road slope, which can be very high in certain roads depending on the mass of the vehicle. This paper aims to quantify the accurate fuel consumption economy when the road slope is integrated or not in the EMS optimization problem of a parallel HEV using dynamic programming.

This article presents a small-scale testbed for automated driving. The main benefits of small-scale testing are safety, affordable hardware, and independence of traffic regulations and test tracks. The testbed using model trucks (scale 1:14) and model cars (scale 1:10) is described in detail and exemplary results using different advanced driver assistance systems are presented. The focus in this work lies on control engineering topics such as trajectory planning, tracking, prediction and estimation algorithms.

Paper 4. Research on adaptive cruise control algorithm considering road conditions, Zhi Yang, Rui He, Sumin Zhang, Jian Wu, Issue: 20 August 2024

As one of the most important ADAS systems, traditional adaptive cruise control is only suitable for relatively simple working conditions such as expressways, and it is difficult to deal with complex traffic environment and road working conditions. For the problem that the traditional cruise control strategy cannot effectively use the terrain to reduce vehicle fuel consumption under slope conditions and when the vehicle enters a small radius curve such as a ramp, the excessively high turning speed will cause the vehicle to skid. This paper proposes an adaptive cruise control system architecture with adaptability to multiple operating conditions. Simulation and real-vehicle test results show that the adaptive control system architecture and optimization algorithm proposed in this paper have good applicability to road conditions, and improve the economy of the system under ramp conditions and safety under curve conditions. With the development of vehicle intelligence and autonomous driving technology, the research object of intelligent transportation system (ITS) has gradually changed from a “person-vehicle” binary independent system to a multiple coupling system based on “person-vehicle-environment-task”. As a branch of ITS, the adaptive cruise control system aims to realize the longitudinal control of the vehicle, reduce the driver's labor intensity during driving, and improve the safety, comfort and economy of the vehicle. The document proposes a cooperative adaptive cruise control (CACC) that employs inter-vehicle wireless communications to safely an adaptive platooning strategy for synchronized merging in the cyclic communication scenario and the strategy is shown in a benchmark merging scenario. In the document the scholars consider a cooperative adaptive cruise control (CACC) system, which regulates inter-vehicle distances in a vehicle string, for achieving improved traffic flow stability and throughput. The document also proposes a CACC technology to automatically regulate the inter-vehicle distances in a vehicle platooning while maintaining the string stability. What is more, the documents have also conducted some research on cooperation through adaptive cruise control and string stability.

However, the studies on CACC are mostly based on transmission delay in network communication, and do not consider the impact of road on vehicle economy and safety. As one of the most important parts of the environment, the road condition directly determines the driving performance of the vehicle and the difficulty of completing the driving task. The ups and downs of the road have a greater impact on the economy of the vehicle, and the fuel consumption level of the vehicle has a significant increase in the slope road section. Therefore, economic issues have become one of the research hotspots of cruise control on sloped road sections. In addition, when the vehicle enters a curve with a small radius such as a ramp or a curve with a low adhesion, the excessively high speed of the corner may cause the vehicle to slip and other dangers. These will all become factors that reduce the performance of ACC. Therefore, optimization for different road conditions has become an important direction to improve the performance of the ACC system.

5. Study on Adaptive Cruise Control Strategy for Battery Electric Vehicle, Issue: 10 Dec 2023

This paper studies the control strategy for adaptive cruise control (ACC) system on a battery electric vehicle (BEV) in the car-following process, and the highlight of this paper is that the regeneration braking of BEV is considered in the car-following process. The hierarchical control structure is adopted for the ACC system. And the structure contains an upper controller and a lower controller. In the upper controller, multiple objectives including the safety, tracking, comfort, and energy consumption are optimized by using the model predictive control (MPC) method. In the lower controller, the energy is recovered during braking. So the energy economy is improved by reducing energy consumption and increasing energy recovery. The proposed ACC strategy is evaluated in simulation experiment. In the simulation experiment, safe tracking for the front vehicle is guaranteed, and the comfort and the energy economy are improved greatly. So the proposed adaptive cruise control strategy can make ACC more widely used in BEVs.

Among various ADAS technologies, the adaptive cruise control (ACC) is an extension of traditional cruise control, which is used to improve the driving convenience and reduce the drivers' mental burden. When there is no vehicle in front of a vehicle with ACC function (own vehicle), the own vehicle is on the state of speed control. When a vehicle (front vehicle) occurs in front of the own vehicle, the own vehicle is on the state of distance control, and the driving process of distance control is called the car-following process in this paper. During the car-following process, the own vehicle can ensure safety by tracking the velocity of front vehicle or keep an expected spacing.

The front-wheel drive BEV is chosen as the modelling object, and the structure of the front-wheel drive BEV is shown in. In the braking process, the motor drives the final drive to apply braking force to the front axle. Since the motor braking is affected by various factors, it is needed to make coordination for the hydraulic braking to complete the braking process. The hydraulic braking is implemented through controlling the hydraulic pressure adjustment unit of the electronic stability control (ESC) system. The vehicle model is designed in Carsem software, but the complete powertrain model of BEV is not included in Carsem, so the external models of motor and battery need to be added. The lithium battery and permanent magnet synchronous motor are selected for BEV in this paper.

6. Adaptive Cruise Control in Electric vehicles with Field Oriented control: by Ana-Maria Petri and Dorin Marius Pretorius published on 14th July,2023.

An adaptive cruise control system is highly used in conventional internal combustion engine vehicles, but due to the electrification of the vehicles, the adaptive cruise control system must be placed in relation with the electric motor control block. This paper presents the relationship between an adaptive cruise control system and the motor control in the case of an electric vehicle. An indirect filed-oriented control of an induction machine is considered for the electric traction system. The adaptive cruise control block computes the required acceleration to ensure the velocity set by the driver based on the set velocity, the actual velocity of the vehicle, a safe distance, and the relative distance between the host car and a leading car. The proposed system was implemented in MATLAB/Simulink and, also, in a low-scale laboratory experimental setup. The correct operation of the system was tested using a speed profile, and both systems, simulated and experimental, have the expected response. The velocity, acceleration, rotor speed, torque response, and relative and safe distance are the responses presented to prove the

The contributions of this paper are the following:

- Illustrates how adaptive cruise control works for an electric vehicle having an induction motor controlled in an indirect field-oriented manner.
- Presents the relationship between an adaptive cruise control system and field-oriented induction machine control system in the case of an electric traction system.
- Tests the efficiency of an electric powertrain system with an induction machine when adaptive cruise control is considered.
- Using a DSPACE platform, a low-scale laboratory prototype system was implemented, allowing the development of a digital field-oriented control.
- The simulation results and experimental ones prove the efficiency of the proposed and implemented system for an electric propulsion system.

7. Multi anticipation for string stable Adaptive Cruise Control and increased motorway capacity without vehicle-to-vehicle communication: by Riccardo Dona, Konstantinos Mattas, Yinglong He, Giovanni Albano, and Biagio Ciuffo published on 14 July ,2023.

Adaptive Cruise Control (ACC) systems have been expected to solve many problems of motorway traffic. Now that they are widespread, it is observed that the majority of existing systems are string unstable. Therefore, small perturbations in the speed profile of a vehicle are amplified for the vehicles following upstream, with negative impacts on traffic flow, fuel consumption, and safety. Increased headway settings provide more stable flow but at the same time it deteriorates the capacity. Substantial research has been carried out in the past decade on utilizing connectivity to overcome this trade-off. However, such connectivity solutions have to overcome several obstacles before deployment and there is the concrete risk that motorway traffic flow will considerably deteriorate in the meanwhile. As an alternative solution, the paper explores multi anticipation without inter-vehicle communication, taking advantage of the recent advancements in the field of RADAR sensing. An analytical study is carried out, based on the most widely used model and parameter settings used to simulate currently available commercial ACC systems, comparing the transfer functions and step responses for the nominal and the multi anticipative formulations. Then, a microsimulation framework is employed to validate our claim on different speed profiles. Analytical results demonstrate that multi anticipation enhances stability without impacting traffic flow. On the contrary, the simulation study indicates that the multi anticipative-ACC can produce higher road capacity even in the presence of external disturbances and for a wide range of calibrated parameters. Finally, optimality conditions for the tuning of the headway policy are derived from a Pareto optimization.

8. Adaptive Cruise Control Strategy Design with Optimized Active Braking Control Algorithm: by Wenguang Wu, Debiao Zou, Jian Ou, and Lin Hu, Published on 21July,2023.

The braking quality is considered as the most important performance of the adaptive control system that influences the vehicle safety and ride comfort remarkably. This research is aimed at designing an adaptive cruise control (ACC) system based on active braking algorithm using hierarchical control. Taking into account the vehicle with safety and comfort, the upper decision-making controller is designed based on model predictive control algorithm. Throttle controller and braking controller are designed with feedforward and feedback algorithms as the bottom controller, where the braking controller is designed based on the hydraulic braking model. The whole model is simulated collaboratively with Amisom, Carsem, and Simulink. By comparison with the full deceleration model, the results show that the proposed algorithm an not only make the vehicle maintain a safe distance under the premise of following the target vehicle ahead effectively but also provide favorable driving comfort.

This paper is aimed at designing a control scheme that could guarantee safety considering the vehicle characteristic and ensure braking comfort at all times. the vehicle longitudinal dynamics model, the hydraulic braking system, and the control mechanism are included in the proposed research model. The main idea of the controller model is as follows:

- The real-time safety distance according to the vehicle speed and the actual distance and relative speed between leader vehicle and follower vehicle are obtained as the controller input.
- The limitation of the acceleration and relative distance of the follower vehicle is calculated by the longitudinal dynamics model.
- The expected acceleration of the follower vehicle is calculated and the optimized brake pressure is transmitted to the executive agency including active brake controller and active throttle controller.
- The braking pressure is produced by the hydraulic braking system, and the vehicle speed slows down. In this process, the brake pressure information is also transmitted to the longitudinal dynamics model.

1.3 Motivation and problem Definition

The primary motivation for developing advanced cruise control systems, like Adaptive Cruise Control is to enhance driver convenience and safety by automating speed adjustments and maintaining a safe following distance from the vehicle ahead. The problem definition focuses on making driving more comfortable and less stressful especially on long journeys or in heavy traffic while also mitigating potential accidents caused by human error in maintaining safe speed and following distance**.**

Motivation:

- Reduced Driver Stress
- Improved Safety
- Enhanced Fuel Efficiency
- Comfortable and Relaxed Driving

Reduced Driver Stress:

- ACC can alleviate driver fatigue by taking over speed control, allowing drivers to focus on other aspects of driving like steering and lane changes.

Improved Safety:

- ACC helps to maintain a safe following distance and potentially preventing collisions or reducing the severity of accidents caused by speeding or inadequate reactions to traffic changes.

Enhanced Fuel Efficiency:

- This helps to maintain the fuel efficient by reducing the speed in the economy so that fuel can be more efficient.

Comfortable and Relaxed Driving:

- The ability to set the desire speed so that it could be very comfortable by adjusting the speed which feels like to be relaxed.

Problem Definition:

Maintaining Safe Following Distance:

- Accurately and reliably adjusting vehicle speed to maintain a safe distance behind the vehicle ahead, especially in varying traffic conditions.

Adapting to Changing Traffic Conditions:

- Ensuring smooth and safe speed adjustments when the vehicle ahead accelerates and decelerates the lanes.

Handling Adverse Road Conditions:

- It helps to maintain the weather conditions like rain, ice barks so that the performance can be ensured in a promising way.

Integrating with Other Systems:

- Taking in a count that integrating ACC with other driver-assistance system such as lane keeping assist and automatic emergency braking.

Addressing Perception Issues:

- Addressing potential driver hesitation and driver in a ACC and providing clear feedback on system status and functionality.

1.4 Objectives:

1.Speed variation according to the speed hump height and displaying it on the LCD display

To design and implement a system that detects the height of speed humps and measures the variation in vehicle speed before and after crossing them displaying the results on an LCD screen in real-time for traffic monitoring and road safety analysis.

2 Collision avoidance.

The primary objective of a collision avoidance system (CAS) is to prevent or mitigate the severity of collisions by alerting drivers. This is achieved through various sensors, such as radar, cameras, and lidar, which detect potential threats and then trigger warnings and take control of the vehicle's braking or steering

3 Increase response time.

To reduce response time in AI-integrated data workflows by optimizing Azure Data Factory pipelines, deploying low-latency ChatGPT endpoints, and implementing efficient prompt engineering and caching strategies.

4 Collision detection.

To design and implement a robust collision detection system that ensures accurate and real-time identification of object interactions within a dynamic environment, enhancing system reliability and performance.

5 To detect the object coming in front of the vehicle by using ultrasonic sensor.

To develop a system that detects objects in front of a vehicle using ultrasonic sensors, enhancing obstacle awareness and improving vehicular safety through real-time proximity sensing.

6 Reduce fuel consumption.

To create and apply innovative methods that decrease fuel usage in transportation systems by enhancing efficiency and optimizing operational practices.

1.5 Scope and Limitations

Scope:

- **Enables vehicles to automatically regulate speed based on traffic flow.**

  The scope of Advanced Cruise Control (ACC) encompasses the development and implementation of systems that enable vehicles to automatically adjust their speed in response to surrounding traffic conditions. By using sensors such as radar, lidar, cameras, and ultrasonic detectors, ACC helps maintain a safe following distance from the vehicle ahead without continuous driver input. It enhances driving comfort, especially during long highway journeys or in stop-and-go traffic, by reducing the need for frequent acceleration and braking. ACC systems also contribute to better fuel efficiency and are foundational technologies for semi-autonomous and autonomous driving. Various forms of ACC exist, including adaptive, stop-and-go, predictive, and cooperative systems, each with increasing levels of intelligence and automation.

- **Improves driver comfort by reducing the need for manual speed adjustments on highways.**

  ACC helps maintain a safe following distance from the vehicle ahead without continuous driver input. It enhances driving comfort, especially during long highway journeys or in stop-and-go traffic, by reducing the need for frequent acceleration and braking. ACC systems also contribute to better fuel efficiency and are foundational technologies for semi-autonomous and autonomous driving. Various forms of ACC exist, including adaptive, stop-and-go, predictive, and cooperative systems, each with increasing levels of intelligence and automation.

  The scope of Advanced Cruise Control (ACC) extends beyond basic speed regulation and encompasses a wide range of intelligent driving support features. ACC is designed to automatically control a vehicle's speed by monitoring the distance and relative speed of vehicles ahead, enabling smoother, safer, and more efficient driving. It plays a crucial role in reducing driver workload during long commutes or in heavy traffic by handling acceleration and braking tasks. Modern ACC systems are integrated with advanced sensing technologies such as radar lidar ultrasonic sensors and computer vision which enable real-time environmental awareness and decision-making. The system is capable of adapting to various driving scenarios including highway cruising, traffic congestion, and even gradual speed adjustments based on road curvature or topography when paired with GPS and map data.

Limitations:

- **Sensor sensitivity:** Performance can be reduced in poor weather conditions like rain, fog, or snow.
- **Limited object detection:** May not detect stationary or suddenly appearing obstacles effectively.
- **Urban challenges:** Struggles in complex traffic environments with unpredictable behaviour (e.g., pedestrians, cyclists).
- **Road condition impact:** Accuracy decreases on sharp curves uneven terrain or roads with unclear markings.
- **Driver dependency:** Requires the driver to remain attentive and ready to take control at any moment.
- **Not fully autonomous:** Cannot handle all driving scenarios, especially emergency or unusual situations.
- **High cost:** Advanced systems can be expensive limiting their use in budget or older vehicles.
- **Risk of overreliance:** Drivers may become overly dependent on the system potentially reducing reaction time

Espite its growing role in modern vehicles, Advanced Cruise Control (ACC) has several limitations that impact its overall effectiveness and reliability. One of the primary challenges is its dependence on sensor accuracy which can be compromised in adverse weather conditions such as heavy rain fog or snow leading to poor detection of vehicles or obstacles. The system also tends to struggle with stationary or suddenly appearing objects as it is primarily designed to track moving vehicles. In complex urban environments where traffic behavior is less predictable and includes pedestrians cyclists and irregular vehicle movements ACC may not respond appropriately increasing the risk of accidents. Sharp curves uneven roads or unclear lane markings can further affect its performance.

Relevance and Type

Relevance: Advanced Cruise Control is important in modern vehicles because it enhances safety reduces driver stress and supports efficient driving. By automatically maintaining safe distances and adjusting speed based on traffic flow ACC helps prevent collisions and promotes smoother travel. It plays a crucial role in the development of intelligent transportation systems and serves as a stepping stone toward fully autonomous driving technologies. Advanced Cruise Control holds significant relevance in the evolution of modern transportation systems. As road traffic continues to increase, the demand for technologies that enhance safety, efficiency, and comfort has become more critical. ACC addresses these needs by automatically adjusting a vehicle is speed based on the distance from the vehicle ahead, reducing the risk of rear end collisions and minimizing human error. It greatly assists drivers during long-distance travel and in congested traffic where maintaining constant attention and speed control can be mentally exhausting. By promoting smoother acceleration and braking patterns, ACC also contributes to better fuel economy and lower vehicle emissions. Furthermore, it serves as a foundational element for more advanced driver assistance systems (ADAS) and is an essential step toward achieving semi-autonomous and fully autonomous vehicles. In this way, ACC is not just a convenience feature but a key technology shaping the future of intelligent and sustainable mobility.

Type: The most basic form is conventional cruise control, which maintains a constant speed set by the driver but does not adjust based on traffic. More advanced is the adaptive cruise control, which uses sensors like radar and cameras to automatically regulate speed and maintain a safe distance from the vehicle ahead. An enhanced version of this is the stop-and-go adaptive cruise control, which is capable of bringing the vehicle to a complete stop and resuming movement in traffic jams, making it ideal for urban or congested driving conditions. Another variation is the predictive cruise control, which utilizes GPS and map data to anticipate road features such as curves, hills, or speed limit changes and adjusts the speed accordingly. The most advanced type is cooperative adaptive cruise control (CACC), which incorporates vehicle-to-vehicle (V2V) communication to share real-time information between cars, allowing for more efficient traffic flow and synchronized driving behavior

Chapter Summary:

Hump detection involves the use of computer vision techniques, such as object detection algorithms like YOLOv8, to identify humps in road images or videos. This process typically includes data collection, preprocessing, training the detection model, and estimating the height of the humps using computer vision techniques. Cruise control, on the other hand, is a system that allows drivers to maintain a steady speed without constantly pressing the accelerator pedal. Traditional cruise control systems maintain a fixed speed set by the driver. Overall, the integration of hump detection and cruise control systems provides a comprehensive solution for managing humps on the road.

CHAPTER 2

METHODOLOGY

Advanced Cruise Controller enhanced with road hump detection involves a series of integrated steps combining computer vision, machine learning, and vehicle control systems. The process starts with collecting a diverse dataset of road images or video clips that contain various types of speed bumps or road humps. This dataset is then carefully annotated to label the humps and specify their respective heights. Using this labeled data the YOLOv8 object detection algorithm is employed to train a model capable of identifying humps in real-time from incoming images or video frames captured by onboard cameras. After successful detection the system extracts the Region of Interest surrounding each hump to analyze its dimensions. A custom speed control algorithm is then designed which takes the estimated height of the hump as input and calculates the optimal vehicle speed to safely and smoothly pass over it. Speed thresholds are defined based on different hump heights to ensure comfort and vehicle stability. Finally, the hump detection and speed control components are integrated into the vehicle's existing cruise control system enabling it to automatically adjust speed according to road surface conditions without requiring driver intervention**.**

2.1 System Design

The system design of the Advanced Cruise Control with integrated hump detection consists of multiple interconnected components that work together to enhance driving safety and automation. The system begins with an image acquisition module, typically a front-facing camera mounted on the vehicle, which continuously captures real-time road images or video frames. These inputs are then fed into an onboard processing unit where the YOLOv8 object detection model identifies and classifies speed bumps or humps. Once a hump is detected, the system isolates the Region of Interest (ROI) around it and estimates its height using image processing techniques or depth sensors, if available. This data is then passed to a speed control algorithm, which determines the appropriate vehicle speed based on the height and type of the detected hump. Predefined speed thresholds ensure the vehicle slows down appropriately before reaching the hump and resumes its previous cruising speed afterward. The final output is sent to the cruise control module, which adjusts the throttle and braking systems accordingly. All components image acquisition, object detection, height estimation, speed control, and cruise integration are synchronized through a central control unit to ensure smooth and responsive vehicle operation in real-time.

2.2 System Integration

System integration involves combining all individual components such as the camera module, object detection algorithm, height estimation unit, and speed control logic into a unified, functioning system within the vehicle. The camera continuously captures real-time road visuals, which are processed by the YOLOv8 model to detect and identify speed bumps. The detected data, including estimated hump height is sent to the control algorithm which calculates the optimal speed adjustment. This decision is then passed to the vehicle is cruise control system enabling automatic speed regulation without driver input. The integration ensures seamless communication between sensors processing units and actuators allowing the entire system to respond in real time for a smooth and safe driving experience. Efficient system integration is critical to ensure accuracy reduce latency and maintain consistent vehicle behavior under varying road conditions.

An Advanced Cruise Control (ACC) system with hump detection refers to the process of bringing together various hardware and software components into a cohesive, real-time operational framework. The integration starts with the vehicle-mounted camera or sensor module that continuously captures road imagery. This raw data is transferred to the onboard processing system where the YOLOv8 object detection model analyses the frames to identify and classify speed humps. Once a hump is detected, additional processing is performed to extract the region of interest and estimate the height or severity of the hump. The information is then passed to a custom-designed speed control algorithm, which dynamically determines the ideal speed based on the hump's characteristics. This speed adjustment decision is communicated to the vehicle's electronic control unit (ECU), which then modifies throttle input or engages braking as necessary, ensuring the vehicle slows down or accelerates smoothly.

All modules image capture detection, processing, decision-making, and actuation must operate in perfect coordination to ensure that responses are timely and accurate. This requires robust communication protocols, often implemented via automotive-grade controllers like CAN (Controller Area Network) or similar real-time systems. Moreover, system integration includes considerations for fault detection, safety overrides, and manual control recovery to ensure the driver can intervene if needed. The overall goal of integration is to create a seamless interaction between perception, processing, and vehicle control that works reliably under real-world driving conditions.

2.3 Normalization

Normalization in the context of an Advanced Cruise Controller refers to the process of standardizing and preprocessing input data to ensure consistent and accurate performance of the system. Since the ACC relies on various inputs such as images from cameras, sensor readings, and speed measurements these data sources may vary in scale, units, or format. Normalization ensures that all input values are adjusted to a common scale, which helps the detection algorithms like YOLOv8 perform more reliably and reduces the risk of errors during object recognition or speed calculation.

- **Standardizes Data Inputs:** Converts data from various sources (camera, sensors, etc.) into a consistent format.
- **Improves Model Accuracy:** Helps detection models like YOLOv8 perform better by minimizing variations in input values.
- **Scales Values Appropriately:** Ensures data like pixel intensities or speed values are within a uniform range (e.g., 0–1).
- **Enables Efficient Training:** Normalized data helps the machine learning model converge faster and more accurately.
- **Enhances Real-time Performance:** Reduces processing time and improves decision-making speed during vehicle operation.
- **Minimizes Sensor Noise:** Smoothens data, reducing the impact of outliers or inconsistencies from sensors.
- **Ensures Smooth System Communication:** Aligns data formats for seamless integration between detection, control, and cruise systems.
- **Supports Safer Driving:** Provides the system with stable inputs, improving the reliability of automatic speed adjustments.

2.4 Description

Advanced Cruise Control is an intelligent driver assistance system that automatically adjusts a vehicle's speed to maintain a safe distance from vehicles ahead. Unlike traditional cruise control, which keeps a constant speed ACC uses sensors such as radar, lidar, or cameras to monitor traffic conditions and adapt the speed accordingly. It enhances driving comfort, especially on highways or in stop and go traffic, by reducing the need for constant acceleration and braking. ACC contributes to road safety minimizes driver fatigue and serves as a foundational technology for semi-autonomous and autonomous vehicles.

An Advanced Cruise Control system refers to the process of converting input data from various sources such as sensors, cameras, and speedometers into a consistent and comparable scale. This step ensures that all inputs are uniformly processed by the detection and control algorithms, such as YOLOv8, leading to more accurate and stable system performance. For example, pixel values from images may be scaled between 0 and 1 and sensor readings may be standardized to reduce the effect of noise or extreme values. Normalization plays a vital role in improving real-time responsiveness supporting efficient model training and ensuring smooth coordination between system components like object detection and speed control.

- **Maintains Safe Distance**: Automatically adjusts vehicle speed to keep a safe gap from the vehicle ahead.
- **Uses Smart Sensors:** Relies on radar, lidar, or cameras to monitor surrounding traffic conditions.
- **Adapts to Traffic Flow:** Increases or decreases speed based on the movement of other vehicles.
- **Enhances Driver Comfort:** Reduces the need for manual acceleration or braking, especially on highways.
- **Supports Stop-and-Go Traffic:** Some systems can handle low-speed traffic, stopping and starting as needed.
- **Improves Road Safety:** Minimizes the risk of rear-end collisions due to consistent distance monitoring.
- **Reduces Driver Fatigue:** Helpful during long drives by automating speed and distance control.
- **Foundation for Autonomy:** Serves as a building block for semi-autonomous and fully autonomous driving systems.

2.5 Requirement Specification

The Advanced Cruise Control system requires several hardware and software components to function effectively. Hardware requirements include sensors such as radar, ultrasonic sensors, lidar, or cameras for detecting vehicles, obstacles, and road features. An onboard processing unit is essential to analyze sensor data and execute control algorithms. A communication interface is needed to connect the ACC system with the vehicle's throttle, brake, and steering controls. Software requirements involve object detection algorithms speed control logic and real-time decision-making modules. Additionally, the system must support data normalization, error handling, and manual override features for safety. All components must work in real time to ensure accurate speed adjustments based on road and traffic conditions.

In addition to basic sensing and processing capabilities, the Advanced Cruise Control system must meet specific functional and performance requirements to ensure reliability and safety. The system should be capable of detecting vehicles and obstacles within a defined range and angle of view, adapting to varying speeds and weather conditions. Real-time processing is essential to allow immediate response to sudden changes in traffic, such as braking or lane changes by other vehicles. The control algorithms must be precise, enabling smooth acceleration and deceleration without causing discomfort. Moreover, the system should include user interface elements that allow drivers to activate, deactivate, or modify settings as needed. To meet industry standards, the ACC must also support diagnostic features, fail-safe operations, and integration with other driver assistance technologies like lane-keeping or emergency braking systems.

2.7 Overview of Technologies

The development and implementation of an Advanced Cruise Control system rely on a combination of modern technologies drawn from the fields of automotive engineering, electronics, and artificial intelligence. Key technologies include radar and lidar sensors for object detection and distance measurement, as well as camera systems that provide visual input for recognizing lanes, vehicles, and road signs. Advanced object detection frameworks like YOLOv8 are used for real-time image analysis and hump detection. These systems are supported by onboard microcontrollers or embedded processors, which handle data processing and decision-making algorithms. Communication protocols such as CAN (Controller Area Network) enable efficient interaction between the ACC system and other vehicle subsystems like throttle control, braking, and steering. In addition, software tools and programming environments are utilized for model training, sensor calibration, and algorithm development. Together, these technologies form an integrated system that enhances driving safety, comfort, and automation.

Algorithms are trained on large datasets to accurately detect and classify road elements such as vehicles, pedestrians, and speed bumps. The integration of real-time operating systems (RTOS) ensures that the system meets the strict timing requirements needed for safety-critical applications. Furthermore, sensor fusion techniques are employed to combine data from multiple sources—such as radar, cameras, and ultrasonic sensors to improve accuracy and reliability in diverse driving environments. Global Positioning System (GPS) and digital maps can also be used in predictive cruise control systems to adjust speed based on road curvature, elevation, or traffic patterns. All these technologies work together to form a robust ACC system that adapts intelligently to real-world driving conditions, offering a smoother and safer driving experience.

2.8 Technologies to be Used

The Advanced Cruise Control system integrates a variety of cutting-edge technologies to ensure intelligent, responsive, and safe driving. Radar and lidar sensors are used for distance measurement and obstacle detection, allowing the system to maintain safe following distances. Cameras support lane detection, traffic sign recognition, and object classification, enabling the vehicle to respond appropriately to its environment. YOLOv8, an advanced object detection algorithm, is implemented for real-time identification of road elements like speed bumps and vehicles. Ultrasonic sensors can provide close-range obstacle detection, especially useful in low-speed scenarios. The system also relies on microcontrollers or embedded processors to process input data and execute control decisions instantly. CAN bus communication ensures seamless interaction between different modules such as the throttle, brakes, and steering systems. Additional support from machine learning models, sensor fusion techniques, and real-time operating systems ensures the ACC system can operate reliably in various road and traffic conditions.

Advanced Cruise Control systems make use of global positioning systems (GPS) and digital mapping to anticipate road conditions such as curves, elevation changes, or upcoming speed zones. These predictive features allow the system to adjust speed in advance, enhancing both safety and fuel efficiency. Artificial Intelligence (AI) algorithms play a significant role in decision-making, helping the system learn from driving patterns and adapt to complex traffic scenarios. Edge computing is also utilized to ensure real-time data processing directly within the vehicle, reducing latency and dependence on external networks. Furthermore, data normalization and sensor calibration techniques are implemented to ensure consistent performance across varying environmental conditions. These combined technologies form the foundation of a highly responsive and adaptive ACC system that brings vehicles closer to full autonomy.

CHAPTER 3

Design and working

3.1 Problem Statement

To develop the advanced brake system in the vehicles and to control the speed of the vehicles according to the humps height on the road so that vehicle speed automatically controlled in cruise control mode.

3.1.1 Basic Idea of the Project

Hump detection systems can vary in complexity and accuracy depending on the sensors and algorithms used. Advanced systems may incorporate multiple sensors, such as radar or ultrasonic, and use sophisticated algorithms to enhance detection accuracy and reduce false positives or negatives.

There were no open speed hump or bump datasets found available in public domain. We have collected images of hump or bump across various locations in and around the institution campus during various time of the day to capture dataset with variety of viewing and illumination conditions. Most of the humps are observed to have poorly maintained with faded paints.

Training and testing of the model is done by using the concept of yolo-v8 with python programming language in google Collab and PyCharm.

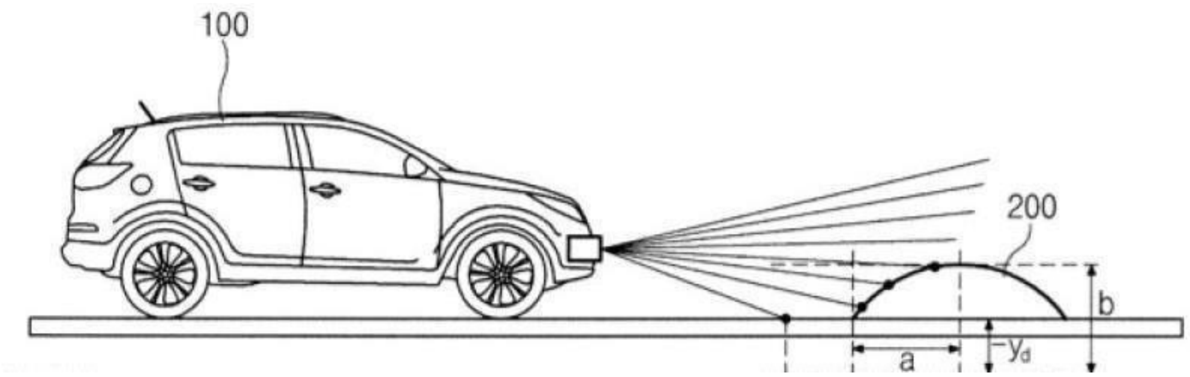

Figure 3.1 Hump detection

The system carefully controls the speed adjustment process, ensuring that the vehicle approaches the speed hump at an appropriate reduced speed. This strategy prevents abrupt changes in velocity, providing a smooth and comfortable ride for the vehicle's occupants

Once the vehicle has safely passed the speed hump, the system gradually restores the original speed or allows the driver to resume control, ensuring a seamless transition. This feature helps maintain the overall flow of traffic and ensures that the vehicle can resume its normal speed without disruption.

3.2 Block Diagram

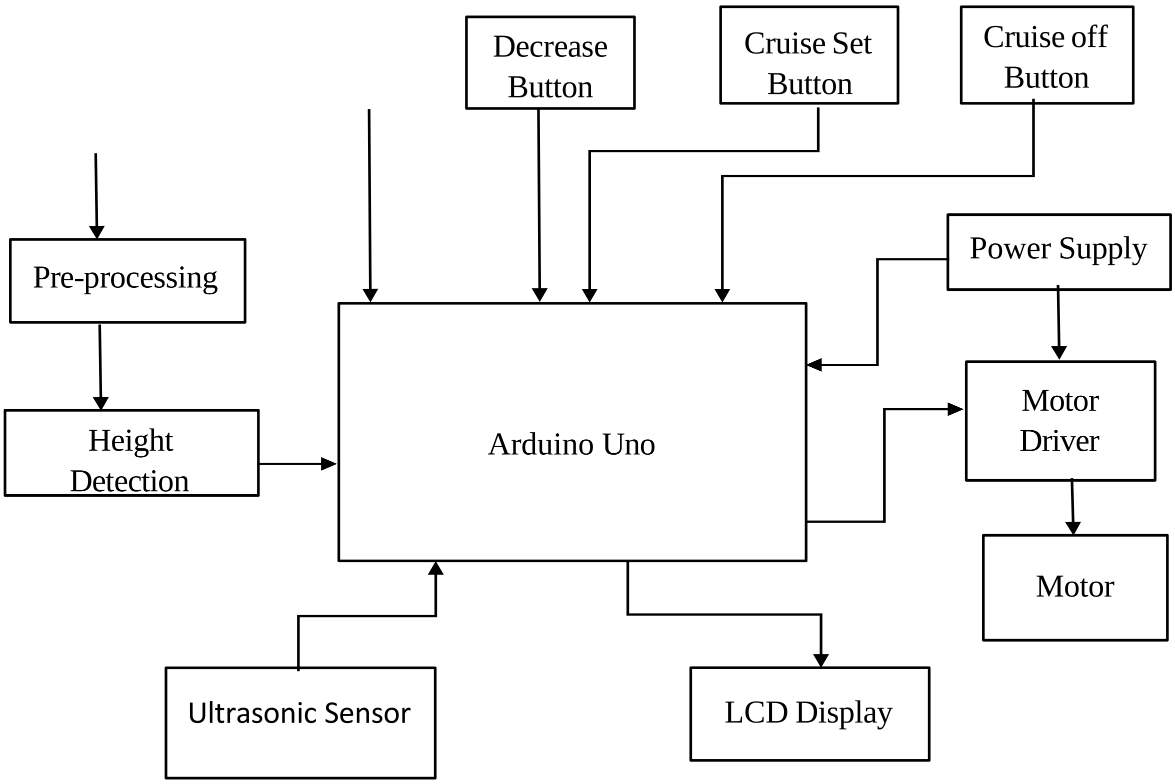

Figure 3.2 Block diagram of ACC

Camera is mounted on the moving vehicle facing towards the road surface and configured to operate at 30fps.Total four buttons are used. Increase button helps in increasing the speed of the motor. Decrease button is used to decrease the motor speed. If we want to set the motor to some constant speed. Then the constant speed is set by pressing the cruise set button. When the vehicle is moving with some constant speed the camera continuously capturing the photos and processes that photos. If any humps are detected then the height will be measured and that height will be sent to the arduino, according the height the arduino sends the signal to the motor driver, the motor driver will control the speed of the motor.

continuously transmits and receives the signal. If any obstacle comes in front of the vehicle ultrasonic sensor sends the signal to the arduino. Arduino sends the signal to the motor driver. Motor driver suddenly stops the motor speed.

3.2 Software Implementation

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <SPI.h>
#include<cvzone.h>
int trigpin=10;
int echopin=9;
long duration;
int distance;
int dir1=7;
int dir2=5;
int enablep=6;
int speed=0;
int rate=0;
int Bp1=4;
int Bp2=3;
int Bp1val;
int Bp2val;
int ccp=2;
int ccpval;
int clrp=1;
int clrval;
```

```
int i;
int min=0;
int set;
int height=0;
LiquidCrystal_I2C lcd(0x27, 16, 2);
SerialData serialData(1,1);
int valsRec[1];
void setup()
{
  serialData.begin(9600);
  pinMode(dir1, OUTPUT);
  pinMode(dir1, OUTPUT);
  pinMode(enablep, OUTPUT);
  pinMode(Bp1,INPUT);
  pinMode(ccp,INPUT);
  pinMode(Bp2,INPUT);
  pinMode(clrp,INPUT);
  digitalWrite(Bp1,HIGH);
  digitalWrite(Bp2,HIGH);
  digitalWrite(ccp,HIGH);
  digitalWrite(clrp,HIGH);
  pinMode(trigpin,OUTPUT);
  pinMode(echopin,INPUT);
  pinMode(13,OUTPUT);
  lcd.begin();
```

```
  lcd.setCursor(0,0);
  lcd.print("Motor Turned On");
}

void loop()
{
  digitalWrite(trigpin,LOW);
  delayMicroseconds(2);

  digitalWrite(trigpin,HIGH);
  delayMicroseconds(10);

  duration=pulseIn(echopin,HIGH);
  distance=duration*0.034/2;
  delay(100);
  Bp1val=digitalRead(Bp1);
  Bp2val=digitalRead(Bp2);
  ccpval=digitalRead(ccp);
  clrval=digitalRead(clrp);
  digitalWrite(dir1, HIGH);
  digitalWrite(dir2, LOW);
  serialData.Get(valsRec);
  height=valsRec[0];
  if(height>0)
  {
```

```
if(height==2)
  min=20;
if(height==4)
  min=15;
if(height==6)
  min=10;
if(height==8)
  min=5;

for(i=rate;i>min;i=i-5)
{
speed=speed-9;
rate=speed*5/9;
lcd.clear();
lcd .setCursor(0,0);
lcd.print("Cruise Mode ");
lcd.setCursor(0,1);
lcd.print(rate);
lcd.setCursor(3,1);
lcd.print("KM/H");
analogWrite(enablep,speed);
delay(500);
}
delay(15000);
increase(rate,set);
```

```
}


if(distance>5 and distance<20)
{
speed=0;
analogWrite(enablep,speed);
lcd.clear();
lcd.setCursor(0,0);
lcd.print("Obstacle found");
lcd.setCursor(0,1);
lcd.print("0");
lcd.setCursor(2,1);
lcd.print("KM/h");
}


if(Bp1val==0)
{
  speed=speed+9;
  rate=speed*5/9;
  Serial.print(rate);
  Serial.println(" KM/h");
  lcd.clear();
  lcd .setCursor(0,0);
  lcd.print("Motor Speed=");
  lcd.setCursor(0,1);
```

```
  lcd.print(rate);
  lcd.setCursor(3,1);
  lcd.print("KM/H");
  delay(100);
}

if(Bp2val==0)
{
  speed=speed-9;
  rate=speed*5/9;
  Serial.print(rate);
  Serial.println(" KM/h");
  lcd.clear();
  lcd .setCursor(0,0);
  lcd.print("Motor Speed=");
  lcd.setCursor(0,1);
  lcd.print(rate);
  lcd.setCursor(3,1);
  lcd.print("KM/H");
  delay(200);
}

if(speed>255)
{
speed=252;
```

```
}

if(speed<0)
{
  speed=10;
}

if(ccpval==0)
{  set=rate;
analogWrite(enablep,speed);
lcd.clear();
lcd.setCursor(0,0);
lcd.print("Cruise Mode Set");
lcd.setCursor(0,1);
lcd.print("to");
lcd.setCursor(3,1);
lcd.print(rate);
lcd.setCursor(6,1);
lcd.print("KM/h");
Serial.print(rate);
Serial.println(" KM/h ");
delay(200);
}

if(clrval==0)
```

```
  {
   for(i=rate;i>0;i=i-5)
  {
   speed=speed-9;
   rate=speed*5/9;
   lcd.clear();
   lcd .setCursor(0,0);
   lcd.print("Cruise Mode off");
   lcd.setCursor(0,1);
   lcd.print(rate);
   lcd.setCursor(3,1);
   lcd.print("KM/H");
   analogWrite(enablep,speed);
   delay(500);
   }
  }
   analogWrite(enablep,speed);
  }


void increase(int pre,int set)
{
 for(i=pre;i<set;i=i+5)
  {
   speed=speed+9;
   rate=speed*5/9;
```

```
lcd.clear();
lcd .setCursor(0,0);
lcd.print("Cruise Mode ");
lcd.setCursor(0,1);
lcd.print(rate);
lcd.setCursor(3,1);
lcd.print("KM/H");
analogWrite(enablep,speed);
delay(500);
}
  }
```

3.4 Flowchart

This is the main idea behind the software implementation. So we have used YOLOv8 particularly for this project. This is the illustration of the entire software implementation right from collecting the data to hump detection.

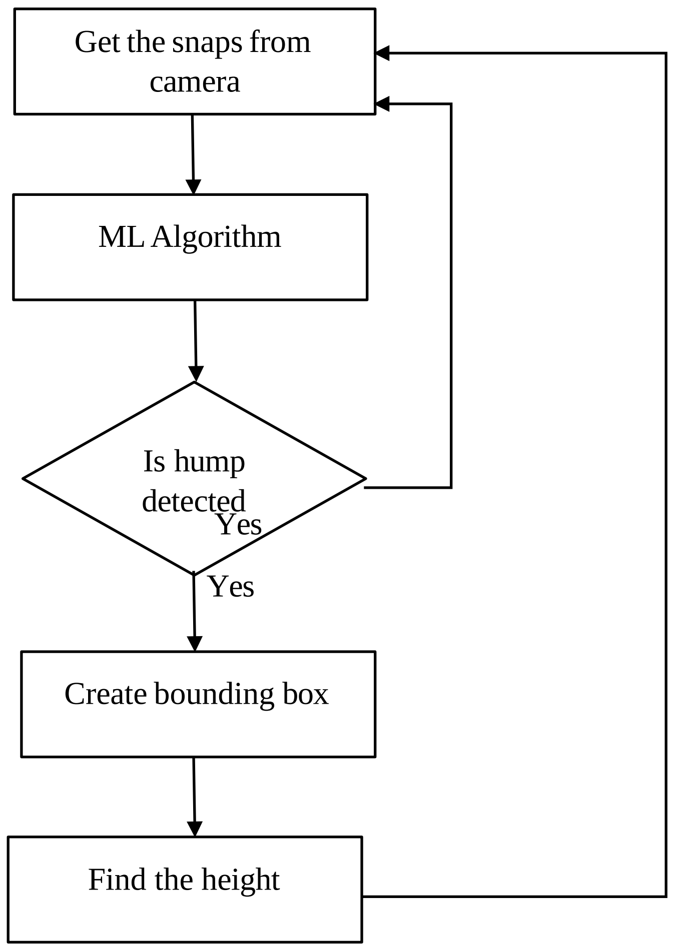

Figure 3.4.1 Flowchart

Flow Chart Description: Camera Captures the images which were used as a data for the machine learning algorithm to train. The data set is divided into two sets for training and testing purpose. For Training we used images and for testing we used the same Hump videos. The captured images were embedded into the machine learning algorithm. Here the YOLOv8 is used for object detection i.e. hump detection the condition here is if the hump detected in the video if yes YOLOv8 creates the bounding boxes and finds the hump height. If the hump not detected it goes to the first step i.e. giving the new data image to the algorithm

3.5 Custom Datasets

Figure 3.5.1 Original image

Figure 3.5.2 Trained image

The above figure 4.4 is the original speed hump image. After training the model we successfully detected the speed hump as shown in the figure 4.5 and the bounding box is created around the hump.

3.5.1Dataset collection

There were no open speed hump/bump datasets found available in public domain. We have collected around 500 images of hump/bump across various locations in and around the institution campus during various time of the day to capture dataset with variety of viewing and illumination conditions. In the dataset collected, we observed the bumps which are having traffic sign and solar cat eyes are very few. Most of the humps are observed to have poorly maintained with faded paints. The dataset collected consists of bumps, humps, rumble strips, shaded bumps due to trees/buildings, bumps with faded paints. A sample of around 250 images from collected dataset is made available public through Mendeley Data.

3.6 Pre-Processing of Data

In order to train an object detection model, we need image height, width, class and bounding box parameters containing xmin, ymin, xmax, ymax of the object. We obtain these by performing labeling. It is performed on our dataset using Labeling, an **open-source** free tool which saves bounding box parameters of each image in to corresponding xml file. All images are compressed in size for faster training on GPUs without losing information that is needed for the algorithm and compromising too much on accuracy.

3.7 Testing of the Model

The rectangular box in the below images indicates the speed hump and corresponding height is also seen. So, we have tested on three conditions.

1. Detection of marked speed hump during day time.
2. Detection of marked speed hump during night time.
3. Detection of unmarked speed hump during day time.

The procedure is that first we have collected the dataset and then by using the labeling we have given a boundary line to the hump, to make sure that hump is detected and these datasets will be dumped in google Collab using latest version of YOLO i.e. YOLOv8 and then a python program is written using PyCharm as an IDE. Giving the path of these datasets in the program we obtained the results as shown below

- **Unmarked speed hump detection:**

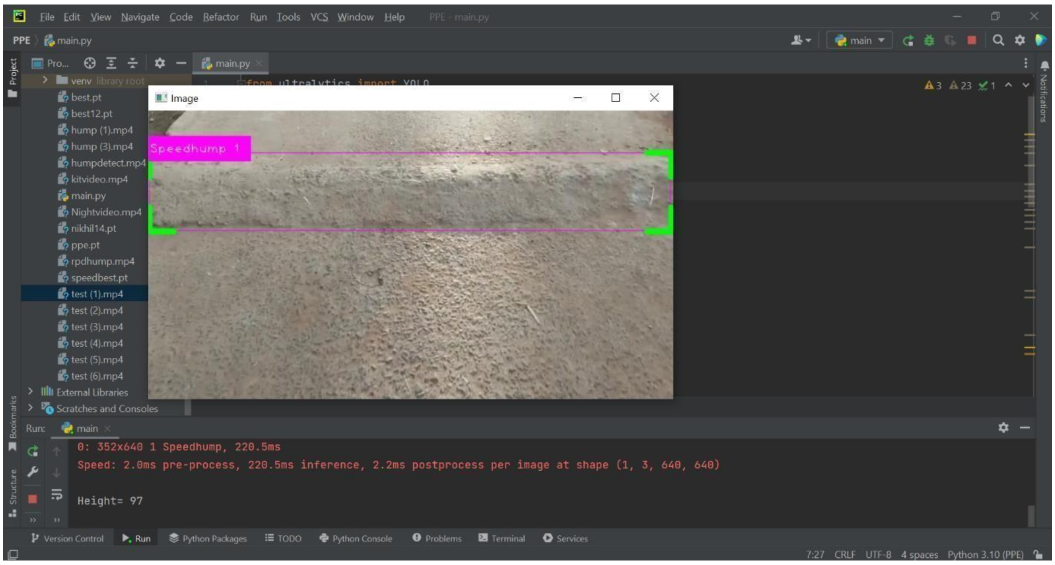

Figure 3.7.1 Unmarked speed hump detection

Camera system captured the footage, we fed the data into the YOLO algorithm for analysis. YOLO's deep neural network architecture analyzed the frames in real-time, scanning for objects that matched the characteristics of a speed hump. The algorithm utilized its comprehensive training on thousands of annotated images to detect the presence of unmarked speed humps accurately.

- **Marked speed hump detection:**

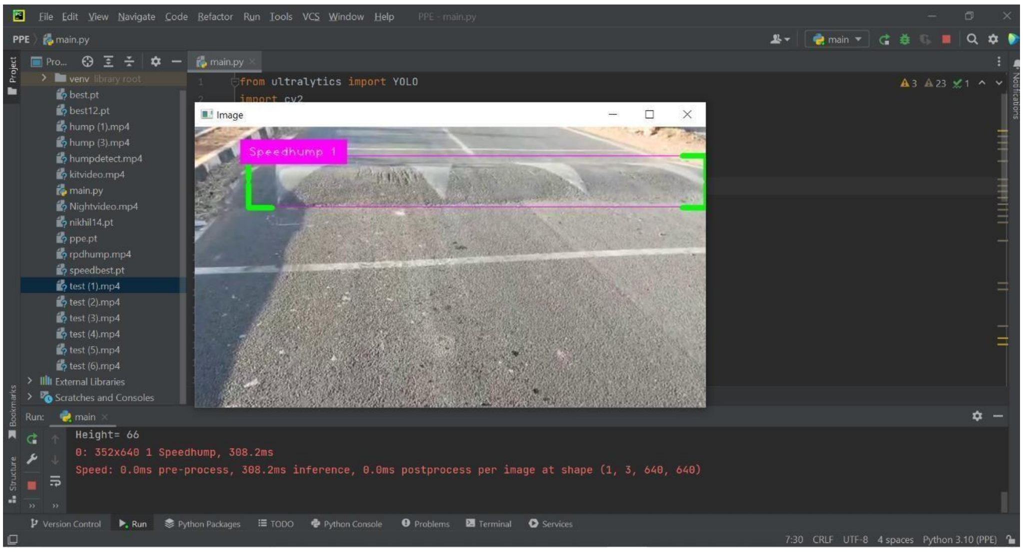

Figure 3.7.2 Marked speed hump detection

- **Marked speed hump detection at night:**

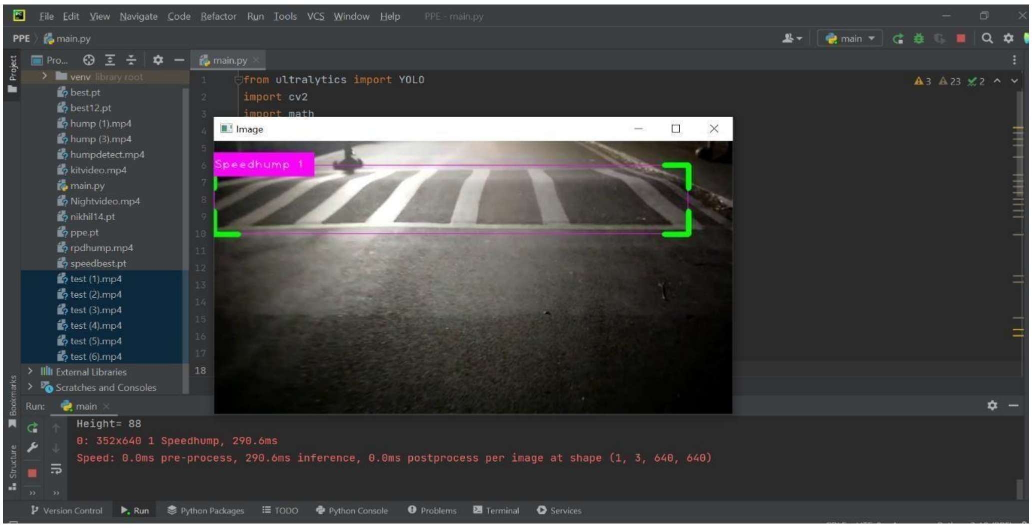

Figure 3.7.3 Marked speed hump detection at night time

From the above figures we can see that whenever the hump is detected the height of the hump will be obtained and from that we can control the speed of the vehicles. So, we have trained during night and day time with both marked and unmarked hump and this program output will be given to Arduino from that we can control the speed. So, this is the basic idea beyond this project, the greater number of datasets will give the more the accurate results.

To get the accurate results we used latest version of yolo which is yolo-v8.

- **YOLOv8**

YOLOv8is the latest version of YOLO, which was launched on January 10th, 2023.A model that defines new state for object detection .in our project we used this to detect road hump. this model introduces developers to a new friendly interface via packages for using this yolo model.so that we can train YOLOv8 model with our custom data and deploy it in minutes. and we can use this model for image classification also. and also, it achieves faster inference speed that any other object detection methods.

Here are few reasons to use YOLOv8

i. YOLOv8 has the highest rate of accuracy measured by coco and Rob flow 100.
ii. It comes with lot of developer convenience features, from an easy to use to a well-structured python package.

3.8 Hardware Implementation

As we have seen first part i.e. software part. Now in the second part we have to implement the hardware part to control the speed and to display it on LCD display. So, the output of the program (hump detection) is given to the Arduino and according to the hump height motor is controlled by the Arduino.

Components Used:

- Arduino uno
- LCD display
- Ultrasonic sensor
- Breadboard
- Motor driver
- Motor

- **Arduino (UNO/Atmel 328)**

The Arduino Uno board is a versatile and widely used microcontroller development board designed for electronics projects and prototyping purposes. It is built around the ATmega328P microcontroller, which provides the processing power and I/O capabilities needed for various applications. The board features 14 digital input/output pins and 6 analog input pins, allowing for the connection of sensors, actuators, and other electronic components. With its built-in USB interface, the Uno board can be easily connected to a computer for programming and communication, eliminating the need for external power supplies in many cases. The board includes a reset button for restarting the microcontroller and putting it into programming mode.

Power can be supplied via USB or an external power source, and a voltage regulator ensures a stable 5-volt supply. The Arduino Uno board is accompanied by a user-friendly Integrated Development Environment (IDE) that simplifies the coding, compilation, and uploading of programs. Its simplicity, versatility, and extensive community support make the Arduino Uno board a popular choice for beginners and experienced users alike, enabling them to bring their ideas to life in the form of innovative projects ranging from robotics to home automation and beyond.

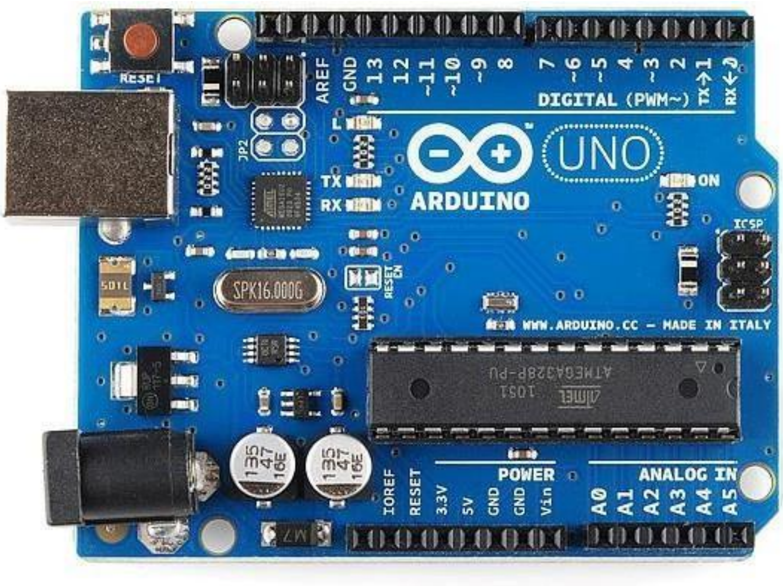

Figure 3.8.1 Arduino uno board

- **LCD Display**

A 16x2 LCD display, also known as a character LCD display, is a commonly used display module in electronics projects. It consists of 16 columns and 2 rows, which allows for the display of up to 32 characters at a time. The LCD display typically uses a parallel interface to communicate with a microcontroller or other devices. It requires a minimum of 6 digital pins for data and control signals, including data lines (D0-D7), a control signal for enabling data transmission (EN), and control signals for selecting between command and data mode (RS) and read or write operations (R/W). The 16x2 LCD display module has a built-in character set, usually consisting of alphanumeric characters, symbols, and some special characters. It allows you to display text, numbers, and symbols in a 2-line format, with each line capable of displaying up to 16 characters.

Figure 3.8.2 16X2 LCD display

- **Ultrasonic Sensor**

Figure 3.8.3 Ultrasonic sensor

Ultrasonic distance sensors are a non-contact devoices to measure the distance between different objects. These types of sensors consist of two parts, one is the transmitter and the other is the receiver, they transmit and receive an ultrasonic sound (Figure 4.11). The basic principle of the ultrasonic sensors is that the transmitter part sends a sound signals with frequency over 18 kHz in the air with speed of 344 m/s (at around 20°C) while the receiver part receives the reflected signals from the target. The distance between the transmitter and the target simply calculated by considering the time required for the ultrasonic signals to reach the receiver from the transmitter, this can be done up to several meters.

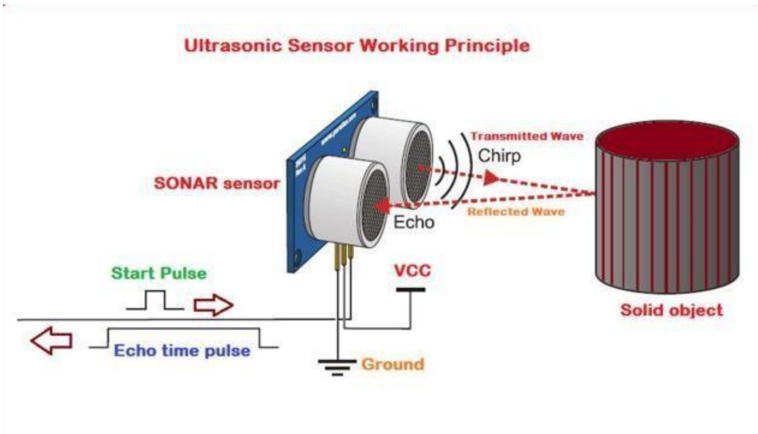

Figure 3.8.4 Ultrasonic sensor working principle

- **Motor driver**

Motor driver is a small type of the electric amplifiers, it receives a control signal with low current and resend it as a signal with higher-current (Yilmaz et al, 2016). The driver is connected to DC-motor of the prototype-car to control the directions and speed of the prototype-car according to PWM. See Figure 4.14.

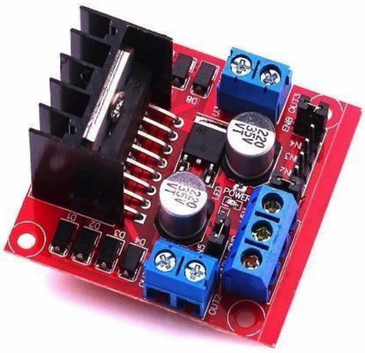

Figure 3.8.5Motor driver

Drive section supply voltage Vs: +5V to +35V.

Drive section peak current Io: 2A.

Logical section of the supply pin area Vss: +5V to +7V.

Logical section of the operating current at range: 0 to 36mA.

- **Motor**

A DC motor is an electric motor that runs on direct current power. In an electric motor, the operation is dependent upon simple electromagnetism. A current-carrying conductor generates a magnetic field, when this is then placed in an external magnetic field, it will encounter a force proportional to the current in the conductor and to the strength of the external magnetic field. It is a device that converts electrical energy to mechanical energy. It works on the fact that a current-carrying conductor placed in a magnetic field experiences a force that causes it to rotate with respect to its original position. Practical DC Motor consists of field windings to provide the magnetic flux and armature which acts as the conductor.

Figure 3.14 DC motor

3.9 Connection of Hardware Components

Connection of Ultrasonic Sensor:

Ultrasonic sensor has 4 pins Vcc pin, Trigger pin, Echo pin, Ground pin. The Vcc pin is connected to 5v supply on bread board and ground pin is connected to ground on the bread board. Echo pin is connected to pin number 9 on the Arduino and trigger pin is connected to pin number 10 of the Arduino board echo pin is input and trigger is output pin.

Connection of Push Buttons:

Push button has 3 pins Signal pin, Ground pin and Vcc. Here 4 push buttons are used increase button, decrease button, cruise mode set button, clear button. The increase button signal pin is connected to pin 1 of the Arduino decrease button is connected to the pin 2 cruise mode signal pin is connected to pin 3 and reset button's signal pin is connected to the pin number 4 of the Arduino board.

Connection of Motor Driver:

Motor driver is connected on the breadboard pin number 1 and pin number 2 of motor driver is connected to the pin number 6 and 7 of the Arduino Uno. Pin number 3 and 6 of motor driver are connected to the DC motor and pin number 4 and 5 of motor driver are connected to the ground and pin number 8 is connected to the battery of 9v.

Connection of LCD Display:

16x2 LCD display is used for printing the data for the errorless output and reducing the number of connecting wires interfaced the I2C with LCD display. LCD display is mounted on the bread board pin1 is connected to the ground of the bread board Pin2 is connected to the Vcc of 9v. Pin number 3 and 4 of Lcd display are connected to analog pin 4 and 5 of Arduino board. Arduino contains everything needed for a control device. Moreover, it can be connected to a computer via a USB cable.

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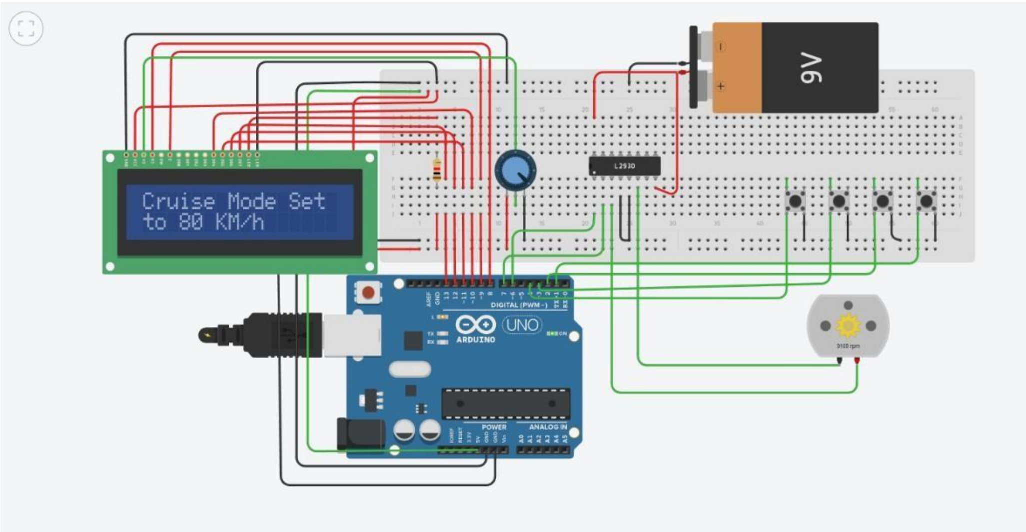

Figure 3.15 Schematic diagram of hardware connection

Summary

In this chapter we have seen the detailed idea of the project and flow of the project and how testing and training is done and the object detection by using YOLOv8 and how the speed is controlled by using Arduino and LCD display. the results and conclusion are discussed in the next chapter.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Results

Once a hump is detected, the system analyses its height, length, and slope to accurately assess its severity. This information allows the system to categorize the hump or assign it a numerical value representing its height. With this data in hand, the system calculates the optimal speed reduction required to safely navigate the hump.

The following are the results obtained after the conduction of this project. figures showing be low are the outputs of both hardware and software part.

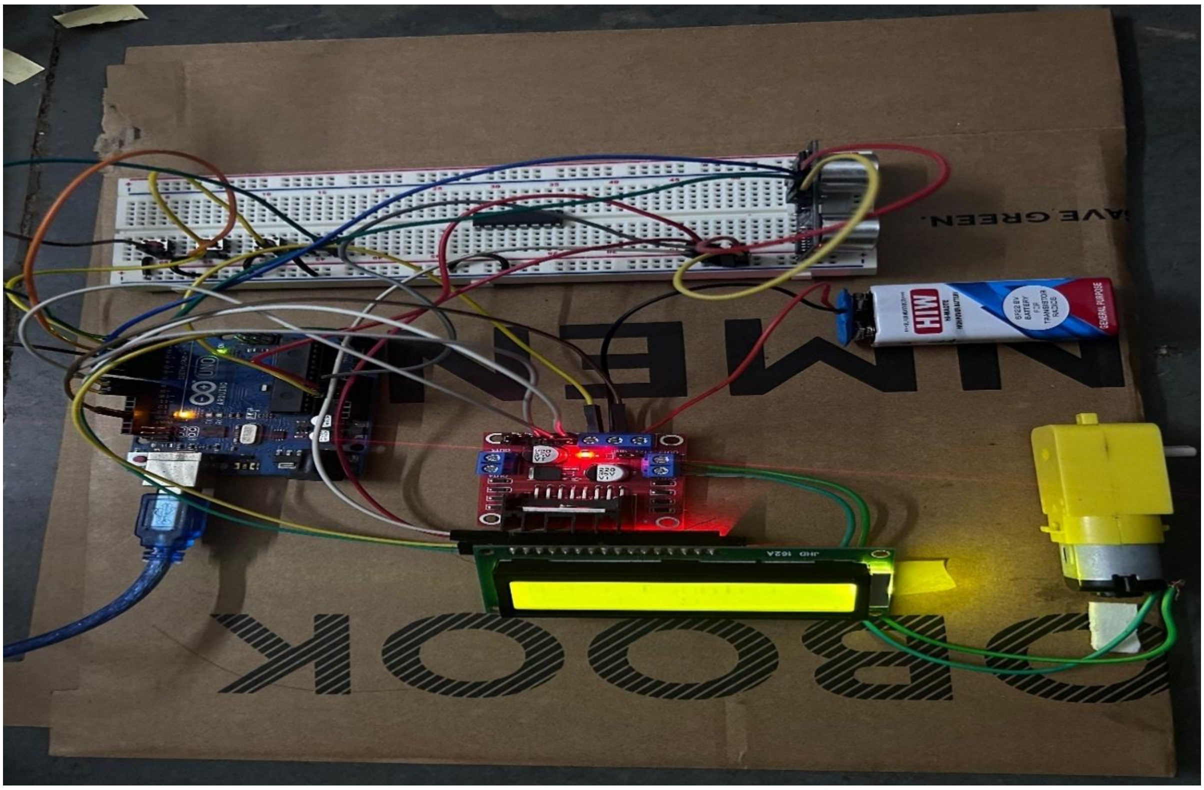

Figure 4.1 Motor speed

Above figure 5.1 show the initially we set the motor speed to 60km/h.

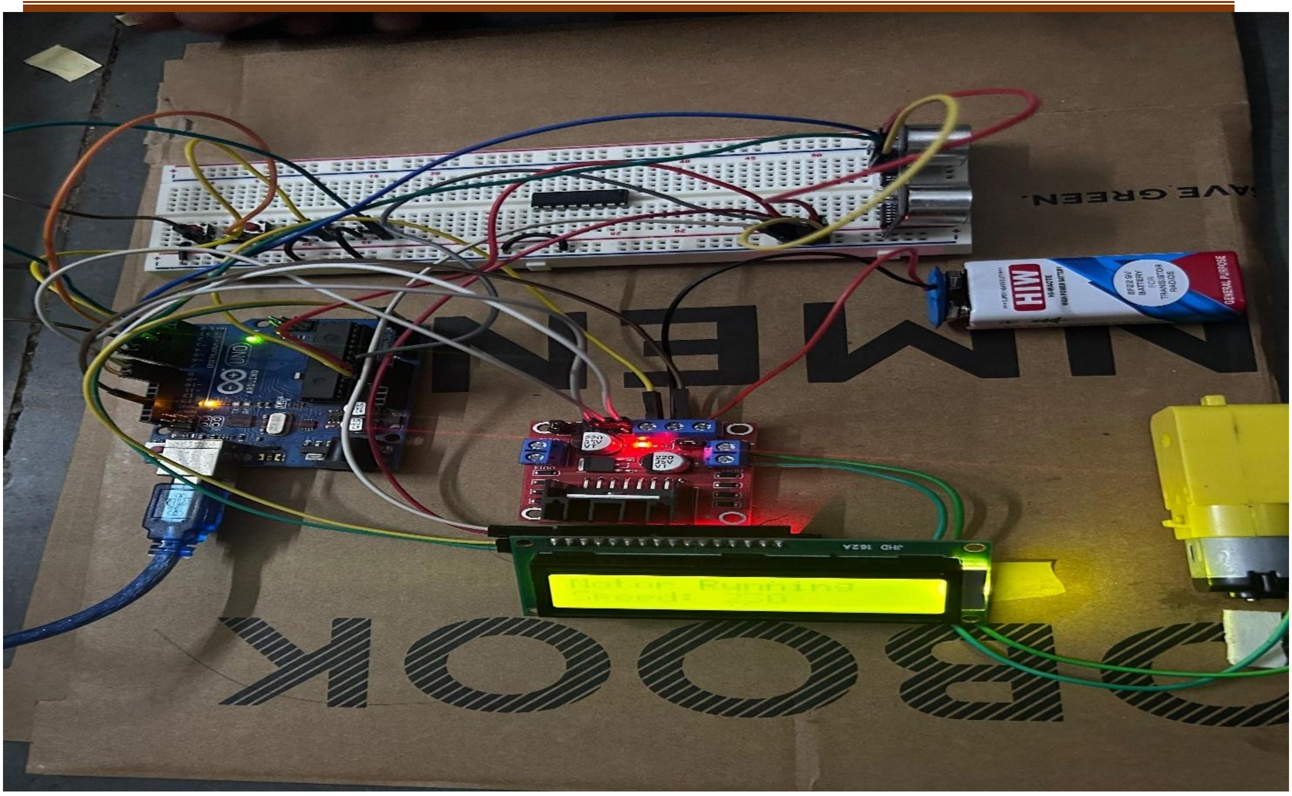

Figure 4.1.2 Setting in cruise mode

Above figure 4.2 shows the setting the motor to cruise mode.

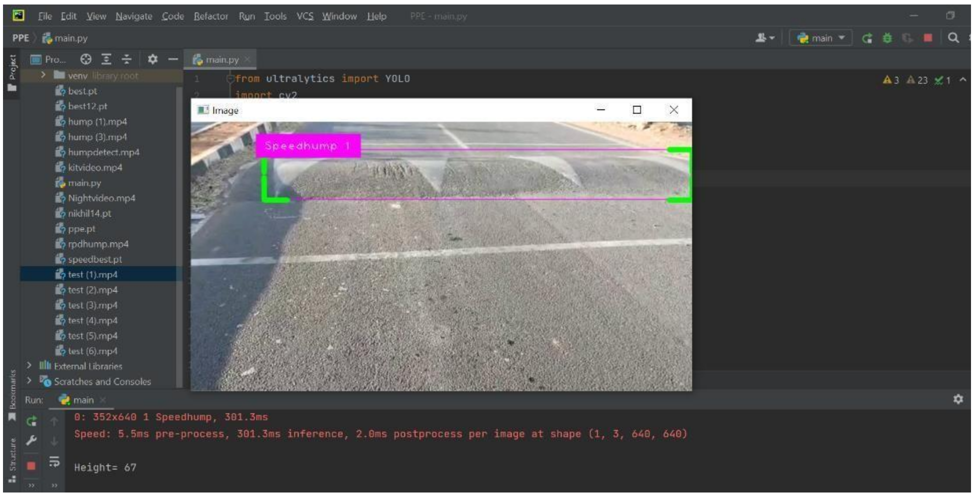

Figure 4.1.3 Speed hump detection

As shown in the figure 4.3 the hump is detected accurately and height is also determined. This height is sent to the hardware.

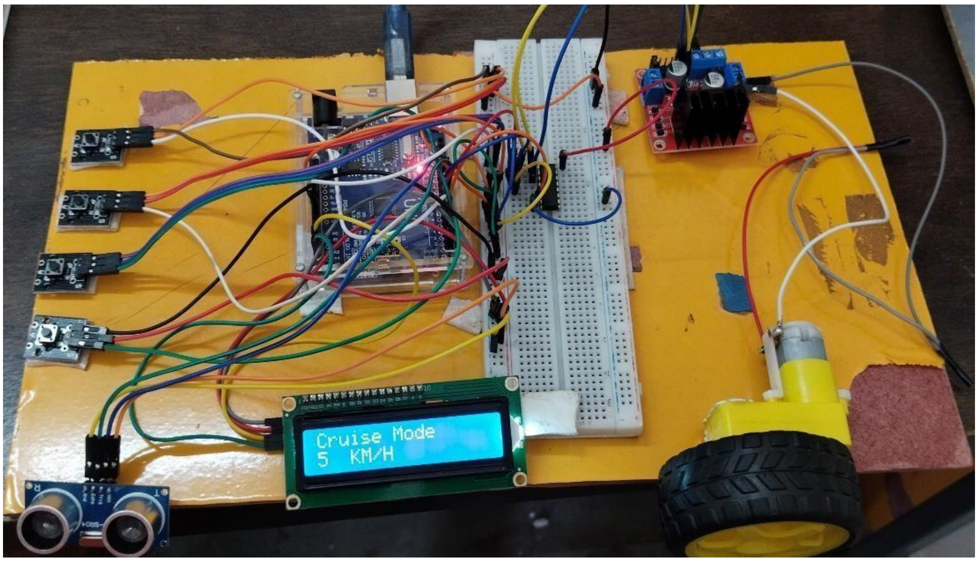

Figure 4.1.4 Speed variation in cruise mode

After hump detection height is send to the Arduino. According to the hump height speed of the motor is controlled by the motor driver.

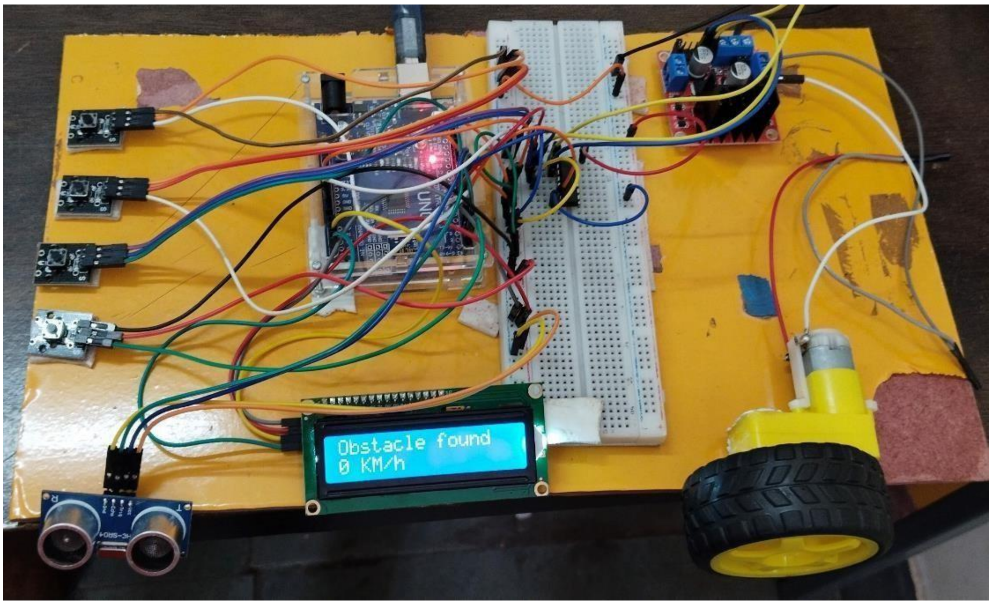

Figure 4.1.5 Obstacle detection

When the any obstacle come Infront of the vehicle it suddenly stops the motor speed as shown in the figure 4.5.

4.2 Performance Analysis

The performance of the Advanced Cruise Control (ACC) system is evaluated based on its responsiveness, accuracy, and ability to maintain safe driving conditions across various scenarios. One of the key performance indicators is how effectively the system maintains a safe following distance in real-time, especially in dynamic traffic conditions. The system's use of sensors such as radar, lidar, and cameras ensures reliable detection of obstacles and vehicles ahead. Additionally, the integration of object detection algorithms like YOLOv8 enhances the recognition of speed bumps and road features, contributing to smoother speed adjustments. The ACC system demonstrates consistent performance in both highway and stop-and-go traffic environments by adjusting vehicle speed appropriately without causing sudden acceleration or braking. Moreover, the system is tested under different weather and lighting conditions to assess its robustness. Overall, the ACC enhances driving comfort, reduces driver workload, and contributes to safer road travel by performing automated speed control with a high degree of precision.

In further evaluating the Advanced Cruise Control system, emphasis is placed on its **reaction time**, fuel efficiency, and **system stability** under real-world driving conditions. The system’s ability to quickly respond to sudden changes—such as a vehicle cutting in or a pedestrian crossing—is critical in preventing accidents. Test results show that the ACC can detect and respond within milliseconds, offering better reaction time than average human drivers. Moreover, by maintaining optimal speeds and reducing unnecessary acceleration or braking, the system contributes to improved fuel economy over long distances. The consistency of performance during lane changes, cornering, and varied road conditions also reflects the system's adaptability and reliability. User experience evaluations indicate increased comfort and reduced fatigue during long drives. These findings collectively highlight the ACC system’s potential to significantly improve driving safety and efficiency without compromising comfort or control.

4.3 Discussion

Accurate hump height detection is crucial for precise speed control and ensuring a comfortable and safe driving experience. By accurately detecting the height of humps, the system can determine the appropriate speed reduction required to safely traverse each hump. This enables the vehicle to adjust its speed in a more precise and controlled manner, minimizing discomfort and the risk of bottoming out or damaging the vehicle. Hump height detection allows for the customization of speed adjustments based on the severity of each hump. The system can categorize humps into different levels or assign numerical values, providing tailored speed reduction recommendations for each hump category. This customization enhances driver comfort and adaptability to different road conditions.

CHAPTER 5
Conclusion and Future Scope

5.1 Conclusion

Advanced Cruise Control (ACC) represents a significant step toward safer and more intelligent driving by combining sensor technologies, machine learning, and automated control systems. It enhances traditional cruise control by dynamically adjusting the vehicle's speed based on real-time traffic and road conditions, ensuring safe following distances and smooth driving experiences. Through the integration of sensors such as radar, cameras, and ultrasonic detectors—along with object detection algorithms like YOLOv8—ACC systems are capable of accurately identifying road features and reacting promptly. The performance analysis confirms that ACC improves fuel efficiency, reduces driver fatigue, and increases road safety. As automotive technology continues to evolve, ACC will play a foundational role in the transition toward fully autonomous vehicles, making everyday driving more reliable, efficient, and comfortable.

Advanced Cruise Control (ACC) has emerged as a transformative driver-assistance technology that combines automation, intelligence, and precision to improve road safety and driving convenience. Unlike traditional cruise control systems, ACC dynamically adapts the vehicle's speed based on traffic flow, road conditions, and the behaviour of surrounding vehicles. By utilizing advanced sensing technologies—such as radar, lidar, ultrasonic sensors, and cameras—along with real-time object detection frameworks like YOLOv8, ACC systems are capable of identifying vehicles, speed bumps, and other road features with high accuracy. These inputs are then processed by intelligent algorithms that control acceleration and braking, ensuring safe following distances and smooth manoeuvring in a variety of driving scenarios, including highways and urban traffic.

The system also contributes to enhanced fuel efficiency by maintaining consistent speeds and minimizing sudden stops or accelerations. Performance analyses show improved reaction times, reduced driver fatigue, and greater adaptability in complex environments, including adverse weather conditions. Furthermore, ACC lays the groundwork for higher levels of vehicle autonomy by integrating seamlessly with other ADAS (Advanced Driver Assistance Systems) technologies. As automotive systems continue to evolve, the capabilities of ACC are expected to expand, leading to safer and more efficient transportation systems. Overall, Advanced Cruise Control represents a crucial milestone in the journey toward fully autonomous vehicles and intelligent mobility solutions.

5.2 Future Scope

The future scope of Advanced Cruise Control (ACC) is vast, driven by rapid advancements in artificial intelligence, sensor technology, and vehicle-to-everything (V2X) communication. Future ACC systems are expected to become more autonomous and context-aware, enabling vehicles to make smarter decisions based on complex traffic scenarios, road conditions, and driver behavior. Integration with real-time traffic data, predictive analytics, and GPS-based mapping will allow ACC to proactively adjust speed not only in response to nearby vehicles but also based on upcoming road features such as curves, inclines, or traffic signals. As 5G connectivity becomes more widespread, ACC systems will also benefit from faster and more reliable communication between vehicles (V2V) and infrastructure (V2I), enhancing situational awareness and coordination.

- **Integration with AI and Machine Learning**
  Future systems will use advanced algorithms to learn from real-world driving data and adapt more intelligently to various driving scenarios.
- **Enhanced Sensor Fusion**
  Combining data from radar, lidar, ultrasonic sensors, and cameras will improve object detection accuracy in all weather and lighting conditions.
- **V2V and V2I Communication**
  ACC systems will communicate with other vehicles and road infrastructure to anticipate traffic flow, signals, and hazards.
- **Predictive Speed Adjustment**
  Using GPS and digital maps, ACC will adjust speed in advance based on road curves, slopes, intersections, and congestion.
- **Better Urban and Low-Speed Adaptation**
  Improved capabilities for navigating in city traffic, stop-and-go conditions, and complex intersections.
- **Fuel and Energy Efficiency Optimization**
  Future ACC will help optimize fuel or battery usage by ensuring smoother driving and reducing unnecessary acceleration.
- **Support for Full Autonomy**
  ACC will evolve into a key component of fully autonomous driving systems minimal human iinput.

- **Smart City Integration**

  ACC systems will contribute to coordinated traffic management, reducing congestion and improving urban mobility.

- **Driver Behavior Prediction**

  Advanced systems may analyze driver behavior to adapt their responses, offering a more personalized and safe driving experience.

- **Cloud Connectivity and Updates**

  Over-the-air updates and cloud-based analytics will keep ACC systems up to date with new features and safety improvements.

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